Business Requirements Document – Mail

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Business Requirements Document (Sub-module: Communication → Mail, code `MAIL`) · **Version:** 1.0 · **Status:** Approved
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
> **Author:** Jojo George (jojogeorge3344@gmail.com)
> © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version
1.0	31-Oct-2026	Jojo George	Approved and signed off; baselined as the agreed business requirements.
1.0	31-Oct-2026	Jojo George	Editorial only: Related Documents paths updated to the `TF - Documents` folder layout; no requirement changed, sign-off unchanged.

Approvals / Sign-off

Role	Name	Decision	Date
Product owner	Jojo George	Approved	31-Oct-2026
Reviewer	Jojo George	Approved	31-Oct-2026

Related Documents

Document	Location
Overall BRD (roles, global rules `BR-nn`, global NFRs, Auth, Legal)	`../../00-Overall/BRD-TaskFlow-Overall.md`
Sibling BRD – Notification (reminder engine that sends reminder mails)	`../04.2-Notification/BRD-Notification.md`
BRD – Reminder Settings ("Also deliver reminders to Mail" toggle)	`../../06-Settings/06.1-Reminder-Settings/BRD-Reminder-Settings.md`
BRD – Task Management (comments, reassignment, worked hours)	`../../02-TaskFlow-Monitoring/02.1-Task-Management/BRD-Task-Management.md`
BRD – Assistant (find user, draft mail)	`../../08-TFA-Assistant/08.1-Assistant/BRD-Assistant.md`

Document	Location
Development Plan (screen-wise) – Screen 14, marked SUPERSEDED	`TF - Documents/Screens Wise Feature Development Plan/14-email-notification-settings-screen.md`
FRD – Mail	`../../../FRD/04-Communication/04.1-Mail/FRD-Mail.md`
HLD / LLD – Mail	`../../../HLD/04-Communication/04.1-Mail/HLD-Mail.md`, `../../../LLD/04-Communication/04.1-Mail/LLD-Mail.md`

Index

1. Project Overview
2. Stakeholders & User Roles
3. Scope
4. Business Process / Workflows
5. Screen-by-Screen Requirements
6. Business Rules
7. Non-Functional Requirements
8. Data Requirements
9. Integrations & External Dependencies
10. Reporting Requirements
11. Recommended Technology Stack (optional)
12. Assumptions, Constraints & Dependencies
13. Risks
14. Acceptance Criteria
15. Requirements Traceability Matrix
16. Project Deliverables
17. Glossary
18. Project Summary

1. Project Overview

1.1 Introduction

Mail is a private internal mailbox inside TaskFlow. Administrators, Managers and Employees use it to read, search, write, reply to, forward and organise messages to each other, save drafts, and send and download attachments. It does not send to outside email addresses. TaskFlow itself also places deadline reminders and task comment alerts into the Inbox as mails from "TaskFlow", so work-related messages are kept in one place.

1.2 Business Problem / Current Situation

Without an internal mailbox, staff discuss task work over outside email or chat tools. Messages are then separated from the tasks they are about, and nobody can see them from TaskFlow. The original plan was to send reminders by outside email (SMTP). That needed server credentials, secret handling

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

and delivery monitoring, which are too heavy for a single-organisation application. It also gave no way for colleagues to write to each other inside TaskFlow.

1.3 Project Objectives

ID	Objective	Success measure
OBJ-MAIL-01	Let users exchange work messages inside TaskFlow without outside email	A sent mail appears in every recipient's Inbox when the sender sees "Mail sent"; outside addresses are refused 100% of the time
OBJ-MAIL-02	Keep every mailbox private to its owner	0 cases where a user can open, flag, delete or download from another person's mailbox; such attempts report "not found"
OBJ-MAIL-03	Make unread mail visible at a glance	The Mail menu badge shows the unread Inbox count, refreshed on every page change and at least every 15 seconds
OBJ-MAIL-04	Support conversations and unfinished work	Reply, Reply all and Forward keep the conversation together; a draft can be saved and continued later
OBJ-MAIL-05	Allow safe file sharing	Limits of 5 files, 10 MB per file and 25 MB per mail are enforced; blocked file types and empty files are refused 100% of the time
OBJ-MAIL-06	Keep task-related system messages next to personal mail	Comment alerts and deadline reminders arrive in the Inbox as mails from "TaskFlow" while "Also deliver reminders to Mail" is on

1.4 Expected Benefits

- **Organisation:** work discussions stay inside TaskFlow, linked to tasks ("Open the task"), with no outside mail server to run.
- **Managers:** can brief team members, reply to questions and pass files on (Forward) without leaving the application.
- **Employees:** one Inbox holds colleagues' mail, task comment alerts and deadline reminders.
- **Every user:** personal folders, stars, important flags and trash; nobody else can read their mail.

---

2. Stakeholders & User Roles

2.1 Stakeholders

Stakeholder	Interest / responsibility
Project owner (Jojo George)	Defines priorities, approves this BRD
Administrator	Grants or removes the `mail.*` permissions for custom roles in Role Configuration; uses Mail like any user
Managers and Employees	Daily users of the mailbox
Owners of related sub-modules (Task Management, Notification, Reminder Settings, Assistant)	Produce or control the system mails that arrive in the Inbox

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

2.2 User Roles

Full role definitions are in the overall BRD. Mail treats every role the same: each user works only on their own mailbox.

Role	Description	Main responsibilities	Access level
Administrator	System administrator	Reads, writes and organises own mail; configures Mail permissions for custom roles	Own mailbox; recipient search over all active users
Manager	Leads a team or project	Reads, writes and organises own mail	Own mailbox; recipient search over all active users
Employee	Works on assigned tasks	Reads, writes and organises own mail	Own mailbox; recipient search over all active users
Custom Role	Role created by an Administrator	As granted: `mail.read`, `mail.compose`, `mail.organize`, `mail.attachments`	Own mailbox, as granted

---

3. Scope

3.1 In Scope

- Mail screen with folders Inbox, Starred, Important, Sent, Drafts and Trash, folder counts and search.
- Reading a mail with its whole conversation; marking it read automatically.
- Unread badge on the Mail menu item.
- Compose panel: new mail, Reply, Reply all, Forward, drafts, recipient suggestions.
- Personal flags: Star, Mark important, Mark unread; Move to Trash, Restore, Delete forever.
- Attachments: upload, remove, forward and download, with size and type limits.
- Receiving system mails: deadline reminders and task comment alerts from "TaskFlow"; reassignment and worked-hours notices from the person who made the change.
- Four permissions: `mail.read`, `mail.compose`, `mail.organize`, `mail.attachments`.

3.2 Out of Scope (this release)

- Sending to or receiving from outside email addresses (SMTP) – replaced by internal Mail (see 3.3).
- SMTP server settings, credentials and "test email" – removed together with outside email.
- Reading another person's mailbox, for any role including Administrator – by design (privacy).
- Formatted (rich text) message bodies – messages are plain text; only web links are made clickable.
- Physically removing deleted mail and attachment files, and automatic clean-up of abandoned uploads – future phase.
- Generating deadline reminders – owned by the Notification sub-module; Mail only receives them.

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

3.3 Modules Overview

#	Module	Purpose	Screens
1	Communication → Mail	Private internal mailbox between TaskFlow users	SCREEN 1 Mail, SCREEN 2 Compose panel

Change from original BRD

Original BRD item	Current requirement
Screen 14 "Email Configuration" (SMTP server, port, sender, username, password, test email)	Superseded and not built. Replaced by the internal Mail screen described here.
Screen 14 FR 1–4 (configure email delivery, secure credentials, never show passwords, test email)	Not applicable: there is no outside mail server and so no credentials.
Screen 14 "Enable/disable email notifications"	Kept as the "Also deliver reminders to Mail" toggle on the Reminder Settings screen.
Screen 14 FR 5 "Send deadline reminders to the appropriate recipients"	Reminders arrive in each recipient's Inbox as mails from "TaskFlow" (sent by the Notification sub-module).
Screen 14 FR 6 "Record email delivery attempts and failures"	Each delivered reminder is recorded in the reminder history (Notification sub-module). Failure tracking is not provided.
Screen 14 FR 7 "Continue in-app notifications if email fails"	Every reminder always creates an in-app notification; mail is an extra copy inside TaskFlow.
Section 3.4 "Email Notifications"	Replaced by internal Mail delivery.

> ⚠️ To be confirmed: original Screen 14 FR 6 and FR 7 assumed an outside mail service that can fail. With internal Mail, a reminder's notification and its Inbox mail are written together; whether a separate failure record is still needed is open.

4. Business Process / Workflows

4.1 Read and Organise Mail

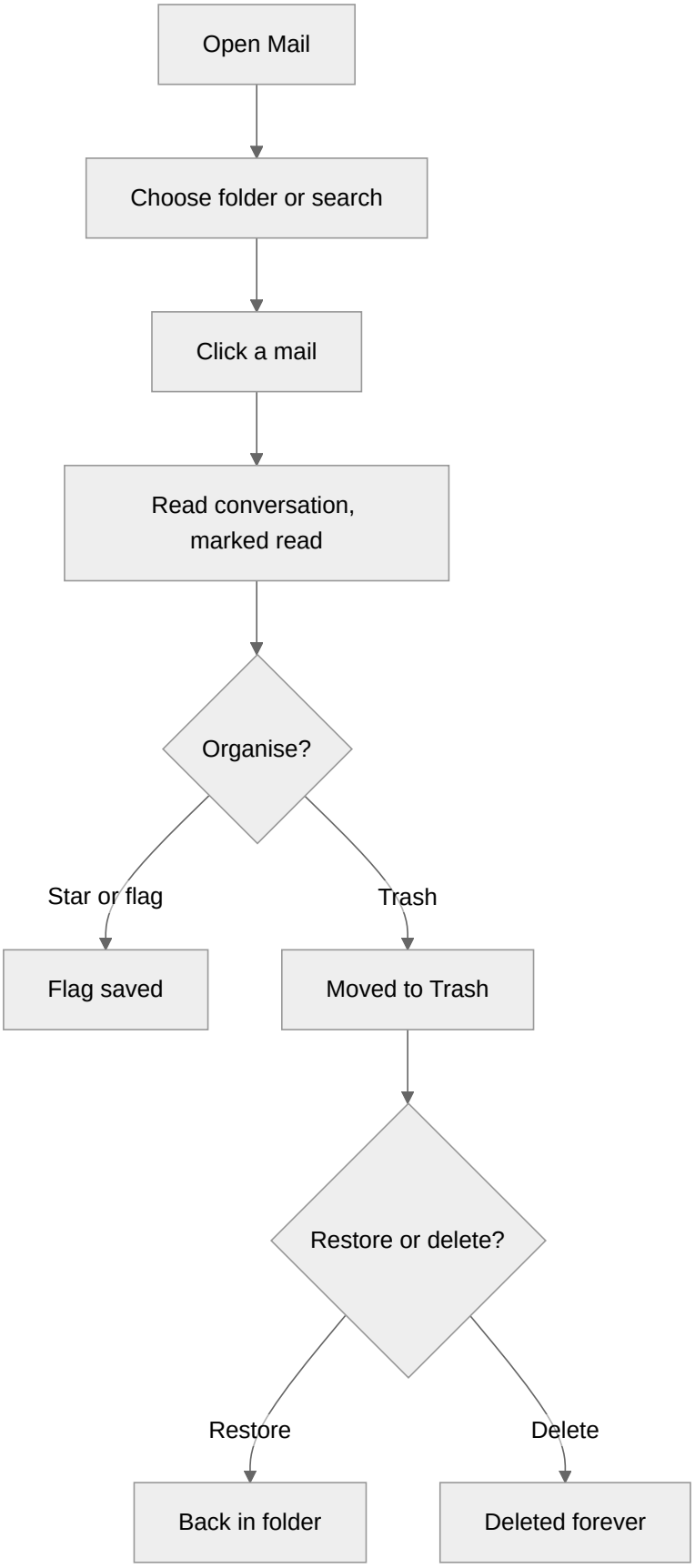
Trigger: User opens Communication Management → Mail · **Actors:** Any user with `mail.read`; `mail.organize` for flags and trash

Step	Actor / System	Action	Result
1	User	Opens Mail	Inbox opens; folder counts shown
2	User	Picks a folder or types in the search box	List shows matching mails, newest first, 25 per page
3	User	Clicks a mail	Reading pane shows the conversation; the mail is marked read
4	User	Stars, marks important, marks unread (Inbox) or moves to Trash	Flag saved; "Moved to Trash" for trash

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

Step	Actor / System	Action	Result
5	User	In Trash, restores or deletes forever	"Restored" or "Deleted forever"
6	System	Updates counts	Folder counts and the Mail menu badge change

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com



Example: Priya (Employee) opens Mail on 31-Oct-2026 and sees "Inbox 3". She opens "Q3 report figures" from Rahul; the count drops to 2. She stars it so it also appears under Starred, then moves an old mail "Lunch on Friday" to Trash.

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

Exceptions / alternate flows:

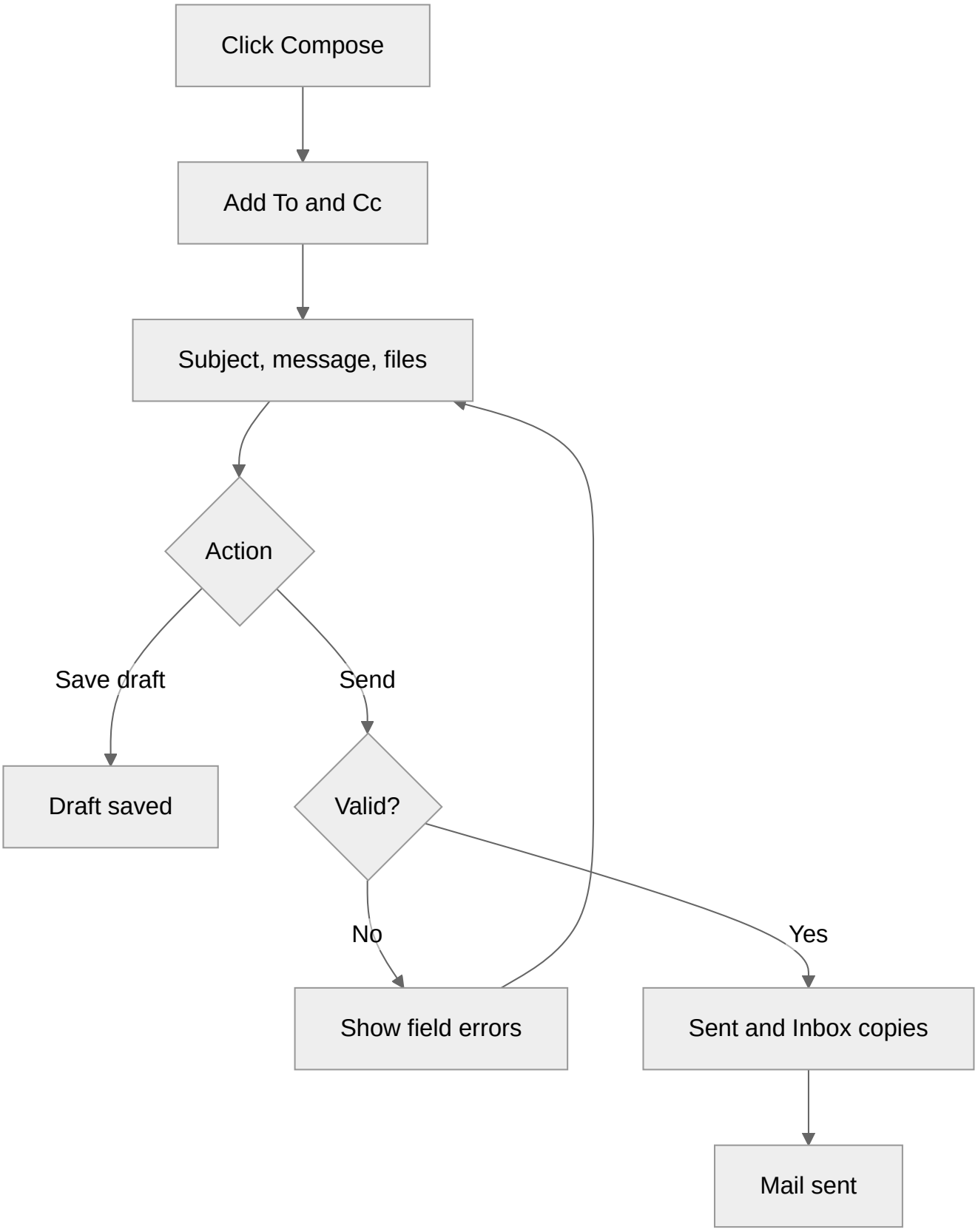
- Folder empty or no search hits: "Nothing in *folder*." or "No mail matches your search."
- The mail belongs to someone else or was removed: "Could not open this mail."
- Delete forever on a mail that is not in Trash: "Move the mail to Trash before deleting it forever."

4.2 Compose and Send a Mail

Trigger: User clicks Compose · **Actors:** User with `mail.compose`; System

Step	Actor / System	Action	Result
1	User	Clicks Compose	Compose panel "New mail" opens
2	User	Types names or emails in To (and Cc)	Up to 10 matching active users suggested; non-users marked "Not a TaskFlow user"
3	User	Enters subject and/or message; may attach files	Files uploaded and shown as chips
4	User	Clicks Send	System checks the mail
5	System	Checks recipients (1–50, all active users), subject or message, lengths, attachments	Errors shown under the field, or the mail is accepted
6	System	Stores the mail	Copy in the sender's Sent and in each recipient's Inbox; "Mail sent"
7	User	Alternatively clicks Save draft or Discard	"Draft saved" or "Draft discarded"

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com



Example: Anita (Manager) writes to Rahul (To) and Priya (Cc), subject "Monthly report – 15 Oct", and attaches "template.xlsx" (220 KB). She clicks Send. Rahul and Priya each see the mail in their Inbox; Anita sees it in Sent.

Exceptions / alternate flows:

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

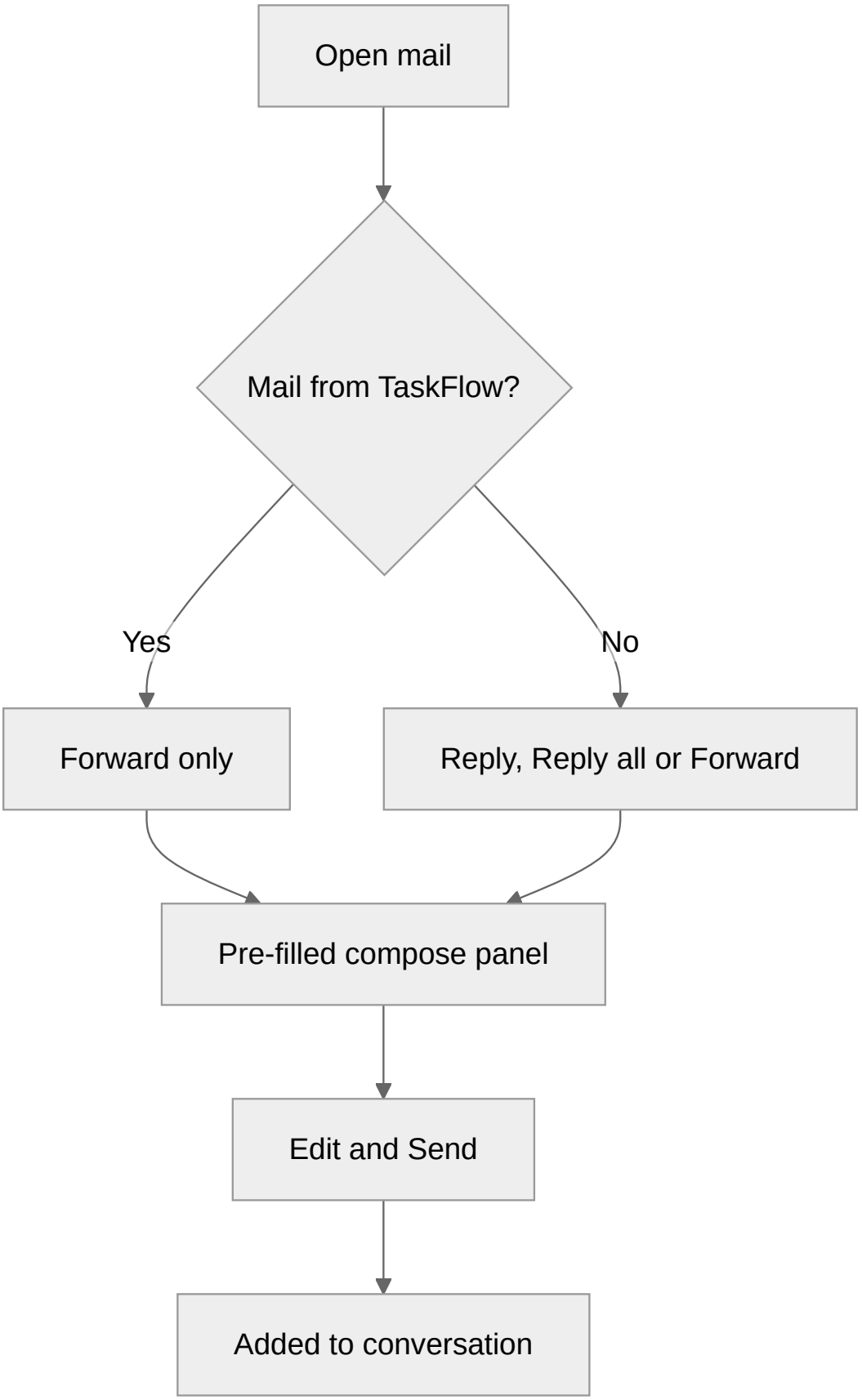
- An address is not an active TaskFlow user: the whole send is blocked – "Not an active TaskFlow user: *addresses*."
- No subject and no message: "Write a subject or a message before sending."
- More than 50 recipients: "A mail can have at most 50 recipients."

4.3 Reply, Reply All or Forward

Trigger: User clicks Reply, Reply all or Forward in the reading pane · **Actors:** User with `mail.compose`

Step	Actor / System	Action	Result
1	User	Opens a mail and picks Reply, Reply all or Forward	Compose panel opens pre-filled
2	System	Pre-fills Reply	Original sender (or original recipients, if you sent it); subject "Re: ..."; quoted "On *date*, *name* wrote:"
3	System	Pre-fills Reply all	Sender plus everyone on To and Cc except you
4	System	Pre-fills Forward	No recipients; subject "Fwd: ..."; forwarded-message header; original attachments
5	User	Edits and clicks Send	New mail joins the same conversation

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com



Example: Rahul replies to Anita's mail. The panel shows To: Anita, subject "Re: Monthly report – 15 Oct" and the quoted text. Anita sees the reply under the same conversation as her original mail.

Exceptions / alternate flows:

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

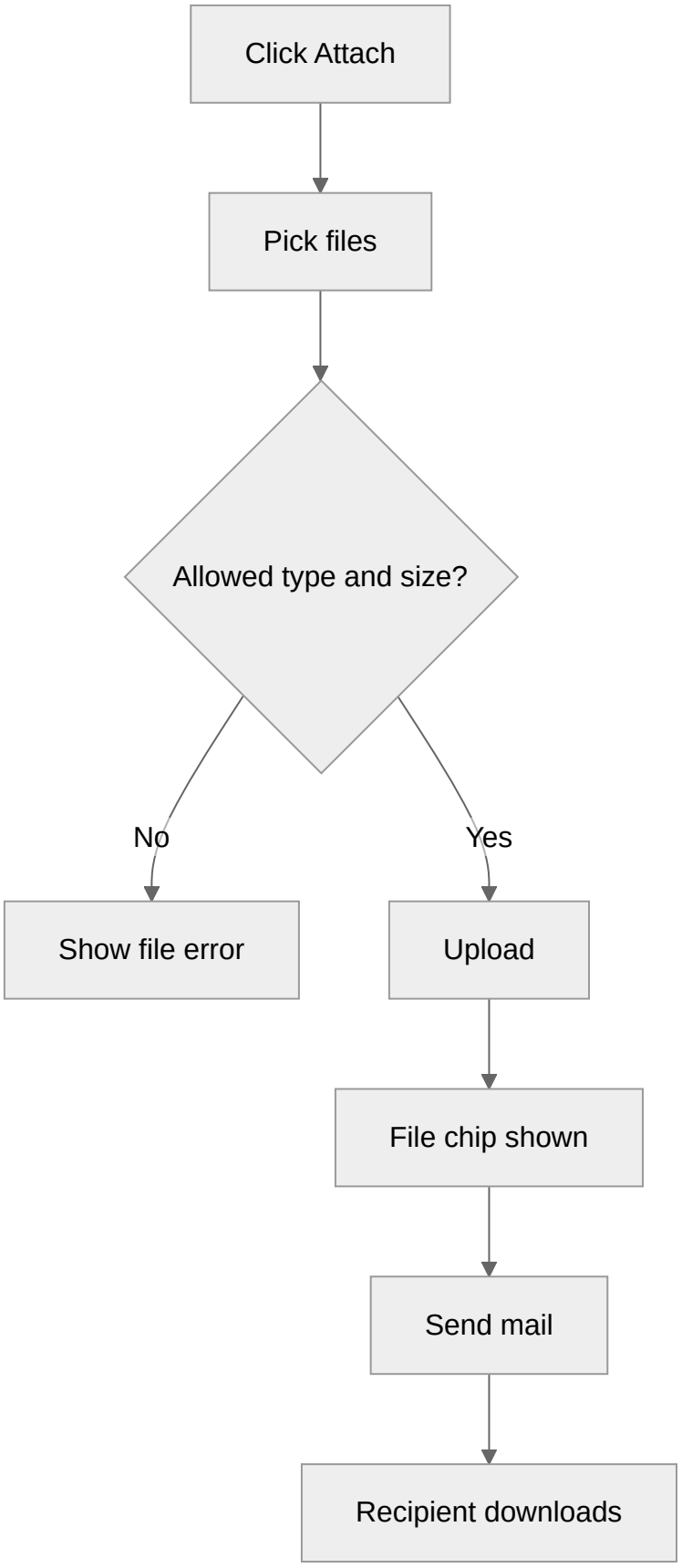
- Reply to a mail from TaskFlow: "Mails from TaskFlow cannot be replied to. Forward it instead."
- Reply and Reply all are not offered for mails in Trash.

4.4 Attach and Download Files

Trigger: User clicks Attach in the compose panel, or an attachment in the reading pane · **Actors:** User with `mail.attachments`

Step	Actor / System	Action	Result
1	User	Clicks Attach and picks files	Files checked for type, empty content and size
2	System	Uploads accepted files	"Uploading…", then a file chip; the user can remove a chip
3	User	Sends the mail	Files go with the mail; forwarded files are copied
4	Reader	Clicks an attachment	File is downloaded (never opened inside the page)

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com



Example: Priya attaches "photos.zip" (12 MB). TaskFlow refuses it: "photos.zip is larger than 10 MB." She attaches "summary.pdf" (3 MB) instead and sends the mail.

Exceptions / alternate flows:

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

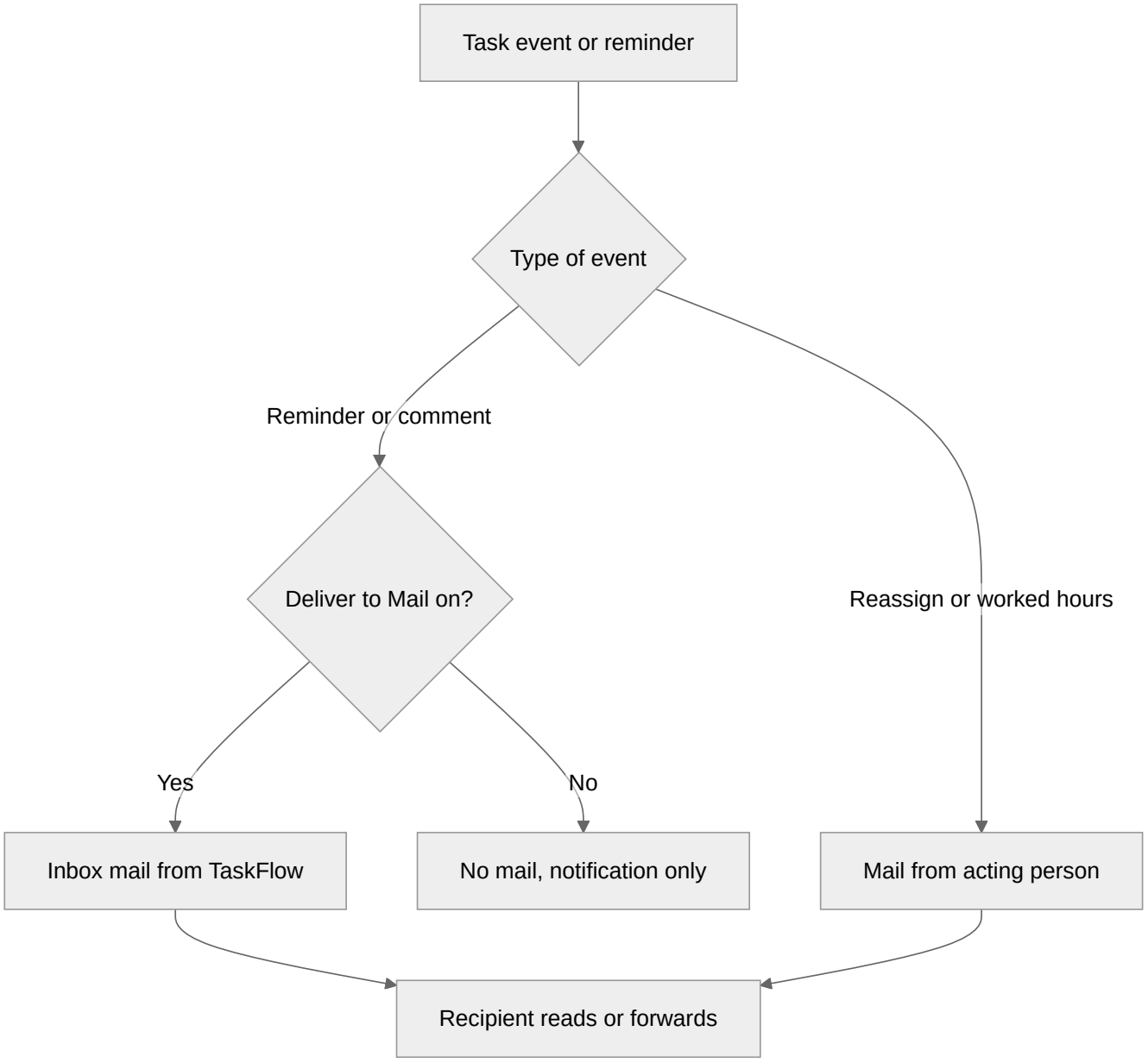
- Blocked type (for example `.exe`): "Files of type .exe are not allowed."
- More than 20 files (or 100 MB) uploaded but not yet sent: "You have *N* files waiting to be attached. Send or discard a mail before uploading more."

4.5 Receive System Mails

Trigger: An event in another sub-module · **Actors:** System; recipients

Step	Actor / System	Action	Result
1	System (Notification)	A deadline, due-today or overdue reminder is delivered and "Also deliver reminders to Mail" is on	Inbox mail from "TaskFlow" with an "Open the task" link
2	System (Task Management)	A comment is added, edited or deleted and the toggle is on	Inbox mail from "TaskFlow", subject "Comment *verb*: *title*"
3	System (Task Management)	A task is reassigned	Mail from the person who reassigned, subject "Task reassigned: *title*"
4	System (Task Management)	Worked hours on a task change	Mail from the person who changed them, subject "Worked hours changed: *title*"
5	Recipient	Opens the mail	Reads it; system mails can only be forwarded

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com



Example: On 13-Oct-2026 Rahul's Inbox receives "Task reminder: Prepare Monthly Report" from TaskFlow. The reading pane offers "Open the task" and Forward, but not Reply.

Exceptions / alternate flows:

- If writing the comment mail fails, the comment itself is still saved; only the mail is dropped.
- Deleted users receive no system mails; reassignment and worked-hours mails go to active users only.

---

5. Screen-by-Screen Requirements

SCREEN 1: Mail

Field	Value
Screen name	Mail (menu: Communication Management → Mail, with unread badge)

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

Field	Value
Module	Communication → Mail
Access	Administrator, Manager, Employee by default; any Custom Role granted `mail.read`
Priority	Must

Purpose Give each user a private place to read, find and organise internal mail, including system mails from TaskFlow.

Screen Elements

- Folder rail: Inbox, Starred, Important, Sent, Drafts, Trash, with counts; Compose button (with `mail.compose`).
- Message list: search box, rows with sender or "To: ...", subject or "(no subject)", preview, date, important flag, paperclip, star toggle; Previous / Next.
- Reading pane: subject, "Open the task" link, conversation with sender, recipients, Cc, date, message, attachment download buttons.
- Reader actions: Reply, Reply all, Forward, Star, Mark important, Mark unread, Move to Trash; in Trash: Restore, Delete forever.
- Narrow-screen layout: one pane at a time with a folder button.

Functional Requirements

ID	Requirement	Priority
REQ-MAIL-01	The system shall open the Mail screen only for users holding `mail.read`, and return other users to their dashboard.	Must
REQ-MAIL-02	The system shall provide the folders Inbox, Starred, Important, Sent, Drafts and Trash, opening Inbox by default.	Must
REQ-MAIL-03	The system shall show the number of unread mails beside Inbox, Starred and Important, and the total number of drafts beside Drafts.	Should
REQ-MAIL-04	The system shall show a badge on the Mail menu item with the number of unread Inbox mails, refreshed on every page change and every 15 seconds.	Should
REQ-MAIL-05	The system shall list the mails of the chosen folder newest first, 25 per page, with Previous and Next.	Must
REQ-MAIL-06	The system shall show on each row the sender ("To: ..." in Sent and Drafts, "TaskFlow" for system mail), the subject or "(no subject)", a preview, the date, an important flag and an attachment indicator.	Must
REQ-MAIL-07	The system shall search the current folder by subject, message text, sender and recipients.	Should
REQ-MAIL-08	The system shall show the whole conversation of an opened mail, oldest first, with sender, recipients, Cc, date, message and attachments.	Must
REQ-MAIL-09	The system shall mark a mail as read for the reader when the reader opens it.	Must
REQ-MAIL-10	The system shall make web links (http and https) in a message clickable.	Could
REQ-MAIL-11	The system shall show an "Open the task" link on a mail that is about a task.	Should

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Requirement	Priority
REQ-MAIL-12	The system shall let users holding `mail.organize` star and un-star a mail, from the list and from the reading pane.	Should
REQ-MAIL-13	The system shall let users holding `mail.organize` mark and unmark a mail as important.	Should
REQ-MAIL-14	The system shall let users holding `mail.organize` mark an Inbox mail as unread.	Should
REQ-MAIL-15	The system shall let users holding `mail.organize` move a mail to Trash and restore it from Trash.	Must
REQ-MAIL-16	The system shall let users holding `mail.organize` delete forever only a mail that is in Trash, or a draft.	Must
REQ-MAIL-17	The system shall apply read state, star, important and trash to the acting user's own copy only.	Must
REQ-MAIL-18	The system shall never show or change another user's mail, and shall report such a mail as not found.	Must
REQ-MAIL-19	The system shall show mails from TaskFlow with the sender "TaskFlow" and offer only Forward for them.	Must
REQ-MAIL-20	The system shall not offer Reply or Reply all for mails in Trash.	Should
REQ-MAIL-21	The system shall keep showing the name of a sender or recipient whose account was later deactivated or deleted.	Should
REQ-MAIL-22	The system shall show one pane at a time on narrow screens, with a button to open the folder rail.	Could
REQ-MAIL-23	The system shall deliver a task comment alert (add, edit, delete) as an Inbox mail from "TaskFlow" to the task's assignee, recipients and the acting person while "Also deliver reminders to Mail" is on.	Should
REQ-MAIL-24	The system shall send a reassignment notice as a mail from the person who reassigned the task to the task's active participants.	Should
REQ-MAIL-25	The system shall send a worked-hours notice as a mail from the person who changed the hours to the task's active assignee and recipients.	Could
REQ-MAIL-26	The system shall let users holding `mail.attachments` download an attachment only when they uploaded it or the mail is in their own mailbox.	Must
REQ-MAIL-27	The system shall always download attachments as files and never open them inside the page.	Must

Validations & Messages

Situation	Rule / message
Mail sent / draft saved / draft discarded	"Mail sent" · "Draft saved" · "Draft discarded"
Trash actions	"Moved to Trash" · "Restored" · "Deleted forever"
Delete forever outside Trash	"Move the mail to Trash before deleting it forever."
Reply to a system mail	"Mails from TaskFlow cannot be replied to. Forward it instead."

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

Situation	Rule / message
Empty folder / no search hits	"Nothing in *folder*." · "No mail matches your search."
Nothing selected	"Select a mail to read it."
Load, open, update or download failure	"Could not load your mail." · "Could not open this mail." · "Could not update the mail." · "Could not mark as unread." · "Could not download the file."
No permission to open the screen	User is returned to their dashboard; no message
No permission for an action	The related "Could not ..." message

Expected Result Each user can find, read and organise their own mail and the system mails about their tasks, and nobody else can see or change their mailbox.

> **Change from original BRD:** this screen replaces original Screen 14 "Email Configuration" (see 3.3).

SCREEN 2: Compose Panel (inside Mail)

Field	Value
Screen name	Compose panel – titled "New mail", "Reply", "Reply all", "Forward" or "Draft"
Module	Communication → Mail
Access	Administrator, Manager, Employee by default; any Custom Role granted `mail.compose` (attachments also need `mail.attachments`)
Priority	Must

Purpose Let users write a new mail, reply to or forward a mail, or continue a saved draft, with recipients checked against the TaskFlow user list.

Screen Elements

- To and Cc recipient pickers with name suggestions; invalid entries marked "Not a TaskFlow user".
- Subject and Message fields.
- Attach button and attached-file chips (removable).
- Buttons: Discard, Save draft, Send.

Functional Requirements

ID	Requirement	Priority
REQ-MAIL-28	The system shall offer Compose, Reply, Reply all and Forward only to users holding `mail.compose`.	Must
REQ-MAIL-29	The system shall suggest at most 10 active users whose name or email matches what the user types in To or Cc.	Should

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Requirement	Priority
REQ-MAIL-30	The system shall mark any recipient who is not an active TaskFlow user as "Not a TaskFlow user".	Must
REQ-MAIL-31	The system shall refuse the whole send when any recipient is not an active TaskFlow user.	Must
REQ-MAIL-32	The system shall require between 1 and 50 recipients in total across To and Cc.	Must
REQ-MAIL-33	The system shall keep a person listed in both To and Cc in To only.	Should
REQ-MAIL-34	The system shall require a subject or a message before sending.	Must
REQ-MAIL-35	The system shall limit the subject to 200 characters and the message to 20,000 characters.	Must
REQ-MAIL-36	The system shall, on send, place the mail in the sender's Sent folder and in each recipient's Inbox, and show "Mail sent".	Must
REQ-MAIL-37	The system shall pre-fill Reply with the original sender (or the original recipients when the user sent the original), the subject "Re: ..." and the quoted text "On *date*, *name* wrote:".	Must
REQ-MAIL-38	The system shall pre-fill Reply all with the sender and everyone on To and Cc except the user.	Must
REQ-MAIL-39	The system shall pre-fill Forward with no recipients, the subject "Fwd: ...", a forwarded-message header and the original attachments.	Must
REQ-MAIL-40	The system shall add a reply or forward to the same conversation as the original mail.	Should
REQ-MAIL-41	The system shall save a draft even when its recipients are incomplete or invalid, and check recipients only when the draft is sent.	Must
REQ-MAIL-42	The system shall let the user open a draft from Drafts and continue it.	Must
REQ-MAIL-43	The system shall remove the files uploaded in the panel when the user discards it.	Should
REQ-MAIL-44	The system shall let users holding `mail.attachments` attach at most 5 files to one mail.	Must
REQ-MAIL-45	The system shall limit each attachment to 10 MB and all attachments of one mail to 25 MB, counting forwarded files.	Must
REQ-MAIL-46	The system shall refuse empty files and the file types `.exe`, `.dll`, `.bat`, `.cmd`, `.com`, `.msi`, `.ps1`, `.sh`, `.js`, `.vbs` and `.scr`.	Must
REQ-MAIL-47	The system shall allow each user at most 20 files, together at most 100 MB, uploaded but not yet attached to a sent mail or draft.	Should

Validations & Messages

Situation	Rule / message
No recipients	"Add at least one recipient."
Recipient not a user (picker)	"Not a TaskFlow user: *addresses*. Mail can only go to people in TaskFlow."

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

Situation	Rule / message
Recipient not a user (on send)	"Remove or fix recipients that are not TaskFlow users." / "Not an active TaskFlow user: *addresses*."
Recipients still being checked	"Checking recipients..."
Too many recipients	"A mail can have at most 50 recipients." / "Add between 1 and 50 recipients."
Invalid address	"'*address*' is not an email address."
Empty mail	"Write a subject or a message before sending."
Too many attachments	"A mail can have at most 5 attachments."
Attachments too large	"Attachments on one mail can total at most 25 MB."
File too large	"*file* is larger than 10 MB." / "A file can be at most 10 MB."
Blocked file type	"Files of type *.ext* are not allowed."
Empty file	"*file* is empty." / "The file is empty."
No file chosen	"Choose a file to upload."
Too many pending uploads	"You have *N* files waiting to be attached. Send or discard a mail before uploading more."
Pending uploads at 100 MB	"Your files waiting to be attached already total 100 MB. Send or discard a mail before uploading more."
Attachment not available	"One or more attachments are not available."
Failures	"Could not send the mail." · "Could not save the draft." · "Could not prepare the reply." · "Could not open the draft." · "Could not discard the draft." · "Could not upload *file*."

Expected Result Users can write, reply to and forward mail to colleagues, attach files within safe limits, and keep unfinished mail as drafts, with clear messages when something is not allowed.

> ⚠️ To be confirmed: the Attach button, file chips and download buttons are shown to every user who can compose or read, even without `mail.attachments`; such users get an error when they use them. Confirm whether these controls should be hidden.
> ⚠️ To be confirmed: Discard on a saved draft needs `mail.organize`. A role with `mail.compose` but without `mail.organize` cannot discard a saved draft. Confirm whether this is intended.

---

6. Business Rules

Global rules that apply here are defined in the overall BRD, section 6 (`../../00-Overall/BRD-TaskFlow-Overall.md`): only authenticated, authorised users can use the system; users must not access another user's private messages; deactivated users cannot sign in; records with dependent data are not deleted in a way that damages history.

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Rule	Applies to
BR-MAIL-01	Mail can only be sent to active TaskFlow users; outside email addresses are never accepted.	SCREEN 2
BR-MAIL-02	Every mailbox is private to its owner; no role, including Administrator, can read another person's mail.	SCREEN 1, SCREEN 2
BR-MAIL-03	Read state, star, important and trash are personal to each copy of a mail.	SCREEN 1
BR-MAIL-04	A mail can be deleted forever only from Trash, or when it is a draft.	SCREEN 1
BR-MAIL-05	Mails sent by TaskFlow (reminders, comment alerts) cannot be replied to, only forwarded.	SCREEN 1, SCREEN 2
BR-MAIL-06	A mail has 1 to 50 recipients across To and Cc; a person in both lists counts once, in To.	SCREEN 2
BR-MAIL-07	A mail needs a subject or a message; the subject has at most 200 characters and the message at most 20,000.	SCREEN 2
BR-MAIL-08	A mail carries at most 5 attachments, each at most 10 MB and together at most 25 MB; blocked types and empty files are refused.	SCREEN 2
BR-MAIL-09	Each user can hold at most 20 files (100 MB) uploaded but not yet attached.	SCREEN 2
BR-MAIL-10	Drafts may hold incomplete recipients; recipients are checked when the draft is sent.	SCREEN 2
BR-MAIL-11	An attachment can be downloaded only by its uploader or by a user who has the mail in their mailbox.	SCREEN 1
BR-MAIL-12	Mail from or to people who were later deactivated or deleted keeps their name.	SCREEN 1
BR-MAIL-13	Reminder and comment mails from "TaskFlow" are delivered only while "Also deliver reminders to Mail" is on; reassignment and worked-hours mails are sent from the acting person.	SCREEN 1

> ⚠️ To be confirmed: reassignment and worked-hours mails are sent even when "Also deliver reminders to Mail" is off. Confirm whether they should follow the same toggle.

7. Non-Functional Requirements

Global NFRs (performance, compatibility, backup, accessibility, maintainability) are in the overall BRD. Sub-module specific requirements:

ID	Category	Requirement	Measure / target
NFR-MAIL-01	Performance	Mail lists are paged so that large mailboxes open quickly	25 mails per page
NFR-MAIL-02	Security	Every mail action works only on the caller's own mailbox	Foreign mail always reported as not found

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Category	Requirement	Measure / target
NFR-MAIL-03	Security	Uploaded files cannot run inside the application	Files always downloaded; blocked types refused; messages shown as plain text with only web links clickable
NFR-MAIL-04	Usability	Errors are shown next to the field that caused them	To, Cc, Message and file errors shown under the field
NFR-MAIL-05	Usability	Mail works on narrow screens	One pane at a time below the wide layout
NFR-MAIL-06	Reliability	A send is all or nothing	Any invalid recipient or attachment blocks the whole send; no partial delivery
NFR-MAIL-07	Usability	Unread mail is visible from every page	Mail menu badge refreshed on page change and every 15 seconds
NFR-MAIL-08	Backup & Recovery	Attachment files are stored outside the database	Backups must include the attachment folder as well as the database
NFR-MAIL-09	Maintainability	Access is controlled by permissions, not by role names	Four `mail.*` permissions editable in Role Configuration

8. Data Requirements

Entity / data set	Purpose	Key fields	Retention / rules
Mail message	One written mail or draft	Conversation, previous mail (for replies and forwards), sender (empty for TaskFlow), system flag, subject, message, related task, related reminder, draft recipients, created / updated / sent time	Kept; a draft has no sent time
Mail recipient	Who a sent mail was addressed to	Mail, user, To or Cc	One entry per user per mail
Mailbox copy	Each user's own copy of a mail in a folder	Mail, owner, folder (Inbox, Sent, Drafts), read, starred, important, trashed time, deleted-forever time	Delete forever hides the copy; it is not physically removed
Mail attachment	A file sent with a mail	Mail (empty while pending), uploader, file name, file type, size, stored file, upload time	Files are kept even after delete forever

9. Integrations & External Dependencies

System / service	Purpose	Required?	Notes
Notification sub-module (reminder engine)	Places deadline, due-today and overdue reminders in the Inbox	Optional	Only while "Also deliver reminders to Mail" is on
Task Management	Comment, reassignment and worked-hours mails; "Open the task" link	Yes	Mail failures do not stop the task action
Reminder Settings	"Also deliver reminders to Mail" toggle	Yes	Owned by the Reminder Settings sub-module

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

System / service	Purpose	Required?	Notes
Application Settings	Application timezone used for dates quoted in replies and forwards	Yes	–
Employee Management	List of active users for recipients and suggestions	Yes	Inactive users cannot receive mail
Assistant (TFA)	Finds users by name and drafts a mail that the user reviews and confirms before it is sent	Optional	Described in the Assistant BRD
Role Configuration	Grants or removes the four `mail.*` permissions	Yes	–
Local file storage	Holds attachment files	Yes	Server folder; no outside email service is used

10. Reporting Requirements

Not applicable for this sub-module.

11. Recommended Technology Stack (optional)

Not applicable for this sub-module. The stack is stated once in the overall BRD.

12. Assumptions, Constraints & Dependencies

Type	Description
Assumption	All correspondents are TaskFlow users of one organisation.
Assumption	Plain-text messages are sufficient for work communication.
Constraint	No outside email: mail cannot leave or enter TaskFlow.
Constraint	Attachment limits: 5 files, 10 MB each, 25 MB per mail; 20 pending files or 100 MB per user.
Dependency	Active-user list from Employee Management.
Dependency	"Also deliver reminders to Mail" toggle from Reminder Settings.
Dependency	Reminder generation from the Notification sub-module.

13. Risks

ID	Risk	Impact	Likelihood	Mitigation
R-MAIL-01	Storage grows because deleted mail and attachment files are never physically removed, and abandoned uploads are not cleaned up	Medium	Medium	Per-user pending-upload limits; plan a clean-up job in a later phase

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Risk	Impact	Likelihood	Mitigation
R-MAIL-02	Recipient suggestions show the names and emails of all active users to anyone who can compose	Low	High	Accept for a single organisation; restrict `mail.compose` for custom roles if needed
R-MAIL-03	Users without `mail.attachments` see attachment buttons that then fail	Low	Medium	Grant the key with `mail.compose`, or hide the buttons (to be confirmed)
R-MAIL-04	A harmful file is shared by mail	High	Low	Blocked file types, size limits, files always downloaded
R-MAIL-05	Users expect mail to reach outside addresses	Medium	Medium	Picker marks non-users "Not a TaskFlow user"; Help explains internal mail

14. Acceptance Criteria

ID	Criterion	Verified by
AC-MAIL-01	A user with `mail.read` opens Mail and sees the six folders with counts; a user without it is returned to the dashboard.	SCREEN 1, REQ-MAIL-01, REQ-MAIL-02, REQ-MAIL-03
AC-MAIL-02	The Mail menu badge shows the unread Inbox count and changes within 15 seconds of a new mail arriving.	SCREEN 1, REQ-MAIL-04
AC-MAIL-03	A folder lists 25 mails per page, newest first, with the row details, and search finds a mail by subject, text, sender or recipient.	SCREEN 1, REQ-MAIL-05, REQ-MAIL-06, REQ-MAIL-07
AC-MAIL-04	Opening a mail shows the whole conversation and marks it read; links are clickable and a task mail shows "Open the task".	SCREEN 1, REQ-MAIL-08, REQ-MAIL-09, REQ-MAIL-10, REQ-MAIL-11
AC-MAIL-05	Star, important, unread, trash, restore and delete forever work only with `mail.organize`, affect only the user's own copy, and delete forever is refused outside Trash.	SCREEN 1, REQ-MAIL-12 – REQ-MAIL-17
AC-MAIL-06	Trying to open or change another user's mail reports "not found".	SCREEN 1, REQ-MAIL-18
AC-MAIL-07	A mail from TaskFlow shows sender "TaskFlow" and offers only Forward; trashed mail offers no Reply.	SCREEN 1, REQ-MAIL-19, REQ-MAIL-20
AC-MAIL-08	Mail from a later-deactivated user still shows that user's name; on a narrow screen one pane is shown at a time.	SCREEN 1, REQ-MAIL-21, REQ-MAIL-22
AC-MAIL-09	With "Also deliver reminders to Mail" on, a new comment creates an Inbox mail from "TaskFlow" for the participants; a reassignment and a worked-hours change create mails from the acting person.	SCREEN 1, REQ-MAIL-23, REQ-MAIL-24, REQ-MAIL-25

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

ID	Criterion	Verified by
AC-MAIL-10	An attachment downloads as a file for its uploader or a mailbox holder, and is refused for anyone else.	SCREEN 1, REQ-MAIL-26, REQ-MAIL-27
AC-MAIL-11	A user with `mail.compose` gets suggestions (at most 10) and sees non-users marked; a send to a non-user is refused as a whole.	SCREEN 2, REQ-MAIL-28 – REQ-MAIL-31
AC-MAIL-12	Sending with 0 or 51 recipients, or with neither subject nor message, or with a 201-character subject, is refused with the listed message; To wins over Cc.	SCREEN 2, REQ-MAIL-32 – REQ-MAIL-35
AC-MAIL-13	A valid send places the mail in the sender's Sent and each recipient's Inbox and shows "Mail sent".	SCREEN 2, REQ-MAIL-36
AC-MAIL-14	Reply, Reply all and Forward are pre-filled as specified and the new mail joins the conversation.	SCREEN 2, REQ-MAIL-37 – REQ-MAIL-40
AC-MAIL-15	A draft with an invalid recipient saves, reopens from Drafts, and is checked only on Send; discarding removes the panel's uploads.	SCREEN 2, REQ-MAIL-41, REQ-MAIL-42, REQ-MAIL-43
AC-MAIL-16	A 6th file, an 11 MB file, a 26 MB total, an empty file, a `.exe` file and a 21st pending upload are each refused with the listed message.	SCREEN 2, REQ-MAIL-44 – REQ-MAIL-47

15. Requirements Traceability Matrix

Objective	Business rule(s)	Screen(s)	Functional requirement(s)	Acceptance criterion
OBJ-MAIL-01	BR-MAIL-01, BR-MAIL-06, BR-MAIL-07	SCREEN 1, SCREEN 2	REQ-MAIL-01, REQ-MAIL-02, REQ-MAIL-05, REQ-MAIL-06, REQ-MAIL-07, REQ-MAIL-22, REQ-MAIL-28 – REQ-MAIL-36	AC-MAIL-01, AC-MAIL-03, AC-MAIL-08, AC-MAIL-11, AC-MAIL-12, AC-MAIL-13
OBJ-MAIL-02	BR-MAIL-02, BR-MAIL-03, BR-MAIL-04, BR-MAIL-12	SCREEN 1	REQ-MAIL-12 – REQ-MAIL-18, REQ-MAIL-21	AC-MAIL-05, AC-MAIL-06, AC-MAIL-08
OBJ-MAIL-03	BR-MAIL-03	SCREEN 1	REQ-MAIL-03, REQ-MAIL-04, REQ-MAIL-09	AC-MAIL-01, AC-MAIL-02, AC-MAIL-04
OBJ-MAIL-04	BR-MAIL-05, BR-MAIL-10	SCREEN 1, SCREEN 2	REQ-MAIL-08, REQ-MAIL-10, REQ-MAIL-20, REQ-MAIL-37 – REQ-MAIL-43	AC-MAIL-04, AC-MAIL-07, AC-MAIL-14, AC-MAIL-15
OBJ-MAIL-05	BR-MAIL-08, BR-MAIL-09, BR-MAIL-11	SCREEN 1, SCREEN 2	REQ-MAIL-26, REQ-MAIL-27, REQ-MAIL-44 – REQ-MAIL-47	AC-MAIL-10, AC-MAIL-16
OBJ-MAIL-06	BR-MAIL-05, BR-MAIL-13	SCREEN 1	REQ-MAIL-11, REQ-MAIL-19, REQ-MAIL-23, REQ-MAIL-24, REQ-MAIL-25	AC-MAIL-04, AC-MAIL-07, AC-MAIL-09

© 2026 Jojo George. All rights reserved. · jojogeorge3344@gmail.com

16. Project Deliverables

#	Deliverable	Format
1	Mail screen and compose panel, with the four `mail.*` permissions	Application source code (repository)
2	Mail data (messages, recipients, mailbox copies, attachments) in the database initialisation scripts	Database migrations
3	BRD, FRD, HLD and LLD for the Mail sub-module	Markdown
4	Automated tests for sending, drafts and conversations, mailbox actions and attachments	Unit and integration tests

17. Glossary

Term	Meaning
Mailbox	A user's own set of folders and mail copies
Conversation	A mail and all its replies and forwards, shown together
Draft	A saved, unsent mail in the Drafts folder
System mail	A mail sent by TaskFlow itself (sender "TaskFlow"), such as a reminder or comment alert
Pending upload	A file uploaded in the compose panel but not yet part of a sent mail or saved draft
Delete forever	Removes a mail from the user's view permanently; only possible from Trash or for a draft
"Also deliver reminders to Mail"	The Reminder Settings toggle that decides whether reminders and comment alerts are also placed in the Inbox

18. Project Summary

Mail gives every TaskFlow user a private internal mailbox that replaces the original outside-email plan. Users read, search, write, reply to, forward and organise mail, keep drafts and share files within safe limits, while system mails about deadlines and comments arrive in the same Inbox. Every action works only on the user's own mailbox, and four permissions let an Administrator decide which roles may read, compose, organise and use attachments.

© 2026 Jojo George. All rights reserved. Unauthorized copying or distribution of this document is prohibited.